

Soumyaratna Debnath

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Research Interests

- Embodied Spatial Intelligence
- Omnidirectional / 360° Scene Understanding
- Neuroscience-Inspired Vision
- 3D Shape Analysis and Scene Understanding

Research and Professional Experience

Aug'25 – Present	Nanyang Technological University (NTU), Singapore <i>PhD Researcher, EmPACT Lab</i> My research focuses on building next-generation perception systems for embodied AI, with an emphasis on bio-inspired visual representations, omnidirectional intelligence, and spatial reasoning. Working with Prof. Addison Lin Wang, I develop efficient vision-language frameworks that enable embodied agents to perceive, and interact with the 3D world more intelligently and efficiently.
Mar'24 to Apr'25	Indian Institute of Technology Gandhinagar, India <i>Researcher, Computer Vision, Imaging and Graphics (CVIG) Lab</i> I conducted research in differentiable rendering, 3D shape analysis, pose estimation, and computer vision. Collaborated with the National Institute of Mental Health (NIMH), USA, L&T Technology Services, the AI Centre of Excellence, and other external research partners.
Aug'22 to Feb'24	Indian Institute of Technology Gandhinagar, India <i>Software Developer, Institute Management System</i> Developed and maintained modules of IIT Gandhinagar's Institute Management System, an institute-wide ERP platform supporting academic and administrative workflows.
Sep'20 to Feb'22	Cooch Behar Government Engineering College, India <i>Research Intern, Department of Computer Science and Engineering</i> Worked on soft-computing and metaheuristic methods for multilevel image segmentation and thresholding; the research resulted in conference publications and a book chapter.

Education

Aug'25 – Present	Nanyang Technological University (NTU), Singapore <i>Ph.D. in Electrical and Electronic Engineering</i> Nanyang President's Graduate Scholar (NPGS), GPA: 4.20/5.00, Supervisor: Prof. Addison Lin Wang
2022 to 2025	Indian Institute of Technology Gandhinagar, India <i>Master of Technology (M.Tech.) in Computer Science and Engineering</i> GPA: 9.67/10, Gold Medal for Outstanding Research, Supervisor: Prof. Shanmuganathan Raman
2018 to 2022	Cooch Behar Government Engineering College, India <i>Bachelor of Technology (B.Tech.) in Computer Science and Engineering</i> GPA: 9.34/10.

Selected Honours and Awards

2026	NeurIPS Spotlight for “AdapToPASS: Ambiguity-aware Adaptive Spherical Transformer for Panoramic Semantic Segmentation.”
2026	CVPR Highlight for “LLMind: Bio-inspired Training-free Adaptive Visual Representations for Vision-Language Models.”
2026	Award of Merit, SRISS 2026 , for presentation of LLMind.
2025	Nanyang President's Graduate Scholarship (NPGS) , Nanyang Technological University.
2025	Gold Medal for Outstanding Research , Indian Institute of Technology Gandhinagar.
2024	Finalist, Qualcomm Innovation Fellowship India , for the proposal “Shadow Guided Optimal Packing and Assembly of General 3D Shapes.”

Peer-reviewed Conference Papers

* equal first authors

- [C1] Soumyaratna Debnath, Weiming Zhang, Shriram Damodaran, Dingwen Xiao, and Addison Lin Wang. **AdapToPASS: Ambiguity-aware Adaptive Spherical Transformer for Panoramic Semantic Segmentation**. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. **Spotlight**.
- [C2] Shriram Damodaran, Soumyaratna Debnath, Cheston Tan, Addison Lin Wang **SphMind: Towards Robust, Training-Free VLM-based Spatial Reasoning with a 360 Camera**. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026, Main Track.
- [C3] Soumyaratna Debnath, Bui Duc Manh, Zinan Liu, and Addison Lin Wang. **LLMind: Bio-inspired Training-free Adaptive Visual Representations for Vision-Language Models**. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3133–3142, 2026. **Highlight**.
- [C4] Soumyaratna Debnath*, Ashish Tiwari*, Kaustubh Sadekar, and Shanmuganathan Raman. **RASP: Revisiting 3D Anamorphic Art for Shadow-Guided Packing of Irregular Objects**. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Main Track, pp. 5849–5858, 2025.
- [C5] Soumyaratna Debnath*, Harish Katti*, Shashikant Verma, and Shanmuganathan Raman. **L3D-Pose: Lifting Pose for 3D Avatars from a Single Camera in the Wild**. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
- [C6] Soumyaratna Debnath, Ashish Tiwari, and Shanmuganathan Raman. **ScribGen: Generating Scribble Art Through Metaheuristics**. In *ACM SIGGRAPH Asia Art Papers*, 2024.

Preprints and Manuscripts

* equal first authors

- [P1] Zinan Liu, Eric Zheng, Soumyaratna Debnath, Hao Shi, Ling Xiao, and Addison Lin Wang. **VL2Spike: Spike-driven Distillation from VLMs for Low-Power Visual Perception in Embodied AI**. *arXiv preprint arXiv:2606.15898*, 2026.
- [P2] Shriram Damodaran*, Soumyaratna Debnath*, Yan Wu, Wei-Yun Yau, and Addison Lin Wang. **EAGOR: Embodied Reasoning in Omni-direction**. *arXiv preprint arXiv:2607.06165*, 2026
- [P3] Soumyaratna Debnath*, Bui Duc Manh*, Zetong Zhang, Shriram Damodaran, Arvind Kumar, Yueyi Zhang, Lu Mi, Erik Cambria, and Addison Lin Wang. **Mind Meets Space: Rethinking Agentic Spatial Intelligence from a Neuroscience-inspired Perspective**. *arXiv preprint arXiv:2509.09154*, 2025.
- [P4] Shashikant Verma, Harish Katti, Soumyaratna Debnath, Yamuna Swamy, and Shanmuganathan Raman. **STEP: Simultaneous Tracking and Estimation of Pose for Animals and Humans**. *arXiv preprint arXiv:2503.13344*, 2025.

Book Chapter

- [B1] Soumyaratna Debnath, Abhirup Deb, Sourav De, and Sandip Dey. **Modified Harris Hawk Optimization Algorithm for Multi-level Image Thresholding**. In *Hybrid Computational Intelligent Systems*.

Leadership and Mentoring

2026 – Present	Academic Coordinator, EmPACT Lab, NTU Singapore. Coordinate academic and research activities within the laboratory.
2025 – Present	Research Mentoring. Mentor master’s students and research interns working on computer vision, multimodal learning, embodied AI, and adaptive visual perception projects.

Selected Research Presentations

Jun 2026	LLMind: Bio-inspired Training-free Adaptive Visual Representations for Vision-Language Models. CVPR 2026, Denver, Colorado, USA.
May 2026	LLMind. SRISS 2026, Singapore. Received the Award of Merit .
Dec 2024	ScribGen: Generating Scribble Art Through Metaheuristics. SIGGRAPH Asia 2024, Tokyo, Japan.